

# Amar Vashishth

Senior Software Engineer

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## Profile

Senior backend and platform engineer with 7+ years building high-throughput commerce systems, messaging infrastructure, Kubernetes services, and internal GenAI engineering tools. Experienced in zero-downtime migrations, production reliability, secure data workflows, and LLM-powered automation. Python, Java, Go, C++, Spring Boot, Kafka, K8s, MySQL, Couchbase

## Education

### Master of Technology | Computer Science and Engineering

2017 – 2019

Indian Institute of Technology Madras

Chennai, Tamil Nadu

- Graduate coursework and research focus on deep learning, pattern recognition, probabilistic methods, Bayesian inference, Markov chains, and linear algebra for ML systems.
- Earned admission through the highly selective IIT entrance process, ranking in the top 0.1 percentile among 500K+ national candidates across 14 computer science subjects.

## Work Experience

### Applications Engineer

July 2023 – Present

Rakuten

Tokyo, Japan

#### – Generative AI Engineering Tools

*Technologies: LLM Agents, Python, Chrome Extension, FastAPI*

- \* **Real-time production log and metrics analysis with AI**, converting noisy operational signals into structured engineering reports.
- \* Built **Chrome extension to integrate LLM assistance** into daily engineering workflows for contextual checks, documentation tasks, and pull-request review support.

#### – High-Scale Commerce Platform Modernization

*Technologies: Java, Spring Boot, Python, Kafka, React, Oracle, Couchbase, Redis*

- \* Modernized browsing-history and gift-history systems, each with several services, supporting **34,000 requests per second** for Rakuten's commerce storefront.
- \* Executed a **zero-downtime, rollback-safe Couchbase-to-Oracle migration** while preserving continuity for high-traffic user-history flows.
- \* Migrated *ashiato.rakuten.co.jp* storefront capabilities to React based frontend.

#### – Cloud Migration, Delivery, and Reliability

*Technologies: Kubernetes, Jenkins, AWS, Istio, Envoy, GSLB, Apache*

- \* Migrated critical commerce services from **bare-metal infrastructure to Cloud-infra with Kubernetes** orchestration, improving deployment consistency and operational scalability.
- \* Served as acting **Tech Lead** coordinating technical priorities, incident response, onboarding, and service continuity.
- \* Improved **Super Sale readiness** through observability enhancements, security hardening, vulnerability remediation, and large-scale load testing.

### Senior Software Engineer

July 2019 – July 2023

Myntra (Walmart)

Bengaluru, India

#### – Messaging, Streaming, and Data Security Platforms

*Technologies: Java, Python, Go, JavaScript, Kafka, Security, React*

- \* Architected and maintained enterprise messaging infrastructure handling **60,000 requests per second** for business-critical workloads.
- \* Built and maintained **SDKs** in Java, Python, JavaScript, and Go to standardize messaging system adoption.
- \* Designed and implemented self-service cross-datacenter Kafka topic replication and secure handling for PII and sensitive data workloads.

## – Business-Critical Backend Systems

*Technologies: Java, Spring Boot, MySQL, REST API Security, C#*

- \* Developed PLM backend services processing 100,000 requests per minute, **replacing PRM360 workflows**.
- \* Built KnowYourClient for real-time API protection and client metadata management across **250K monthly requests** without service modifications.
- \* Delivered a **C# Excel automation plugin** that reduced manual validation and upload effort for PLM business users.

## – Reliability, Observability, and Platform Operations

*Technologies: ELK Stack, Prometheus, Grafana, Docker, MySQL 8, Load Testing*

- \* Integrated ELK logging, StatsD, InfluxDB, Prometheus, and Grafana with email and Slack alerting for mission-critical services.
- \* Containerized backend services, **led MySQL 5.7 to MySQL 8 migration** work, and performed **Kafka brokers stress tests** before major sale events.

## Projects and Research

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### Deep Learning Thesis: Nutritional Analysis of Indian Cuisines

*Technologies: CNNs, Computer Vision, TensorFlow, Data Engineering, Rule-Based Reasoning*

- Built a CNN workflow to identify Indian dishes from images and support automated nutritional analysis.
- Estimated calories by combining model outputs with deterministic nutrition rules and rule-based reasoning.
- Created Indian-dish datasets through government-agency coordination, web crawling, and label cleanup.

### Advanced Pattern Recognition & Deep Learning Academic Projects

*Technologies: Python, TensorFlow, PyTorch, NumPy, SciPy, Scikit-Learn*

- Designed and evaluated neural architectures including Sequence-to-Sequence (Seq2Seq) Networks, text embeddings with word2vec, time series data, and image classification via Fashion MNIST using CNNs etc.
- Built and analyzed probabilistic architectures including Bayesian Neural Networks, Restricted Boltzmann Machines (RBMs), Hidden Markov Models (HMMs), and Gaussian Mixture Models (GMMs) for advanced pattern identification tasks.

### Probabilistic ML and Robotics Academic Projects

*Technologies: Bayesian Inference, Markov Chains, SPARQL, Java, Gmail API*

- Implemented SPARQL/RDF query systems and a Java-based Bayesian email classifier integrated with the Gmail API.
- Engineered autonomous robot navigation systems utilizing real-time obstacle avoidance rules.

## Skills

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**Languages:** Java, Python, Go, JavaScript, TypeScript, C#, C/C++, Rust

**Backend and Web:** Spring Boot, FastAPI, React, Node.js, REST APIs, Nginx, Apache HTTP Server

**AI and Machine Learning:** Proficient Using LLM Agents like Codex, Claude, Gemini; MATLAB, TensorFlow, PyTorch, CNNs, RNNs, LSTMs

**Data and Messaging:** Kafka, RabbitMQ, Redis, MySQL, Oracle, MongoDB, Couchbase

**Cloud and DevOps:** Kubernetes, Docker, Jenkins, GitHub Actions, Ansible, AWS, Azure

**Reliability and Testing:** Prometheus, Grafana, ELK Stack, StatsD, InfluxDB, Gatling, k6, JUnit, Mockito